Two Production Lines, Two Sampling Plans

Unit 4 · Topic 4.1 · THE PROBLEM

Suppose Lines A and B each contain 1,000 cycle times with population mean 40 seconds and standard deviation 12 seconds. Line A is approximately normal and an SRS of $n_A = 9$ is selected; Line B is strongly right-skewed and an SRS of $n_B = 36$ is selected. An auditor wants $P(\bar{X} > 46)$ for each.

Serena: "Only B can use a normal model because $36 \geq 30$; its probability is 0.00135. A is too small."

Mateo: "Only A can use a normal model because its population is normal; its probability is 0.0668. B is too skewed."

Sampling plans

Line	Shape	N	n	μ	σ
A	Normal	1000	9	40	12
B	Right-skewed	1000	36	40	12

FIRST TAKE — one sentence before you work the parts

Which plan seems more likely to give averages close to 40 seconds? Give one reason from the sample sizes or shapes.

DOK 1 **(a)** State the mean of each sampling distribution and the 10% condition for using $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.

DOK 2 **(b)** Verify the 10 percent condition, compute both sampling-distribution standard deviations, and calculate each normal tail probability when justified.

DOK 3 **(c)** In 3–4 sentences, explain what each student gets right and wrong. Use the two normality pathways to decide which probabilities are supported. Connect the different tail probabilities to the sampling SDs in (b).

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.